The Project MARS curriculum consists of 21 lessons designed to be implemented over a 6 week instructional period. The following chart outlines each lesson and how it aligns with Minnesota state standards (http://education.state.mn.us/MDE/EdExc/StanCurri/K-12AcademicStandards/index.htm) and Common Core State Standards (http://www.corestandards.org/the-standards/mathematics/introduction/standards-for-mathematical-practice/).

SBI Lesson Objectives	Common Core Standards	MN State standards
Lesson 1: Ratios Students define a ratio as a multiplicative relationship. They identify the base quantity for comparison of quantities involving part-to-part and part-to-whole. Common Core Standards: 6.RP.1, 6.RP.3, 7.NS.2.b, 7.RP.2.a MN State Standards: 6.1.2.1, 7.1.1.1, 7.1.1.5, 7.1.2.5	**6.RP.1:** Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities. *For example, "The ratio of wings to beaks in the bird house at the zoo was 2:1, because for every 2 wings there was 1 beak." "For every vote candidate A received, candidate C received nearly three votes."* **6.RP.3:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. **7.NS.2.b:** Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers; Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. Interpret quotients of rational numbers by describing real world contexts. **7.RP.2.a:** Recognize and represent proportional relationships between quantities; Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin.	**6.1.2.1:** Identify and use ratios to compare quantities; understand that comparing quantities using ratios is not the same as comparing quantities using subtraction. **7.1.1.1:** Know that every rational number can be written as the ratio of two integers or as a terminating or repeating decimal. Recognize that π is not rational, but that it can be approximated by rational numbers such as 22/7 & 3.14. **7.1.1.5:** Recognize and generate equivalent representations of positive and negative rational numbers, including equivalent fractions. **7.1.2.5:** Use proportional reasoning to solve problems involving ratios in various contexts. *For example: A recipe calls for milk, flour and sugar in a ratio of 4:6:3 (this is how recipes are often given in large institutions, such as hospitals). How much flour and milk would be needed with 1 cup of sugar?*
Lesson 2: Ratios Students use visual diagrams to understand the meaning of equivalent	**6.NS.4:** Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple	**6.1.1.6**: Determine greatest common factors and least common multiples. Use common factors and common multiples to calculate with fractions and find equivalent fractions.

ratios. They identify integer ratios in their lowest or simplest form. They correctly determine if ratios are in simplest form by using division of common factors. Common Core Standards: 6.NS.4, 6.RP.3, 7.NS.2.b, 7. RP.2.a MN State Standards: 6.1.1.6, 6.1.2.4., 7.1.1.1, 7.1.1.5, 7.1.2.5	of a sum of two whole numbers with no common factor **6.RP.3:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. **7.NS.2.b:** Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers; Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. Interpret quotients of rational numbers by describing real-world contexts. **7.RP.2.a:** Recognize and represent proportional relationships between quantities; Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin.	**6.1.2.4:** Use reasoning about multiplication and division to solve ratio and rate problems. *For example: If 5 items cost $3.75, and all items are the same price, then 1 item costs 75 cents, so 12 items cost $9.00.* **7.1.1.1:** Know that every rational number can be written as the ratio of two integers or as a terminating or repeating decimal. Recognize that π is not rational, but that it can be approximated by rational numbers such as 22/7and 3.14. **7.1.1.5:** Recognize and generate equivalent representations of positive and negative rational numbers, including equivalent fractions. **7.1.2.5:** Use proportional reasoning to solve problems involving ratios in various contexts. *For example: A recipe calls for milk, flour and sugar in a ratio of 4:6:3 (this is how recipes are often given in large institutions, such as hospitals). How much flour and milk would be needed with 1 cup of sugar?*
Lessons 3 and 4: Solving Ratio Word Problems Students apply ratio concepts to solve word problems. They represent information in the problems using a ratio diagram and plan to solve the problem using various solution strategies**.** Common Core Standards: 6.RP.3, 7.RP.2.a MN State Standards:	**6.RP.3:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. **7.RP.2.a:** Recognize and represent proportional relationships between quantities; Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin.	**6.1.2.4:** Use reasoning about multiplication and division to solve ratio and rate problems. *For example: If 5 items cost $3.75, and all items are the same price, then 1 item costs 75 cents, so 12 items cost $9.00.* **6.1.3.5:** Estimate solutions to problems with whole numbers, fractions and decimals and use the estimates to assess the reasonableness of results in the context of the problem. *For example: The sum 1/3 + 0.25 can be estimated to be between 1/2 and 1, and this estimate can be used to check the result of a more detailed calculation.* **7.1.1.5:** Recognize and generate equivalent representations of positive and negative rational numbers, including equivalent fractions. **7.1.2.5:** Use proportional reasoning to solve problems involving ratios in various contexts. *For example: A*

6.1.2.4, 6.1.3.5, 7.1.1.5, 7.1.2.5		*recipe calls for milk, flour and sugar in a ratio of 4:6:3 (this is how recipes are often given in large institutions, such as hospitals). How much flour and milk would be needed with 1 cup of sugar?*
Lesson 5: Rates Students define rate as a comparison of two quantities with different units. They understand and learn how to calculate unit rates. Students learn to solve problems in which they compare two rates. Common Core Standards: 6.RP.2, 6.RP.3.b, 7.RP.1, 7.RP.2.b MN State Standards: 6.1.2.4, 7.2.2.1, 7.2.2.2, 7.2.2.3	**6.RP.2:** Understand the concept of a unit rate *a*/*b* associated with a ratio *a:b* with $b \neq 0$, and use rate language in the context of a ratio relationship. *For example, "This recipe has a ratio of 3 cups of flour to 4 cups of sugar, so there is 3/4 cup of flour for each cup of sugar." "We paid $75 for 15 hamburgers, which is a rate of $5 per hamburger."* **6.RP.3.b:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations.; Solve unit rate problems including those involving unit pricing and constant speed. *For example, if it took 7 hours to mow 4 lawns then at that rate, how many lawns could be mowed in 35 hours? At what rate were lawns being mowed?* **7.RP.1:** Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. *For example, if a person walks 1/2 mile in each 1/4 hour, compute the unit rate as the complex fraction 1/2/1/4 miles per hour, equivalently 2 miles per hour.* **7.RP.2.b:** Recognize and represent proportional relationships between quantities; Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.	**6.1.2.4:** Use reasoning about multiplication and division to solve ratio and rate problems. *For example: If 5 items cost $3.75, and all items are the same price, then 1 item costs 75 cents, so 12 items cost $9.00.* **7.2.2.1:** Represent proportional relationships with tables, verbal descriptions, symbols, equations and graphs; translate from one representation to another. Determine the unit rate (constant of proportionality or slope) given any of these representations. *For example: Larry drives 114 miles and uses 5 gallons of gasoline. Sue drives 300 miles and uses 11.5 gallons of gasoline. Use equations and graphs to compare fuel efficiency and to determine the costs of various trips.* **7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.2.3:** Use knowledge of proportions to assess the reasonableness of solutions. *For example: Recognize that it would be unreasonable for a cashier to request $200 if you purchase a $225 item at 25% off.*
Lessons 6 and 7: Solving Proportion Word Problems Students apply ratio/rate	**6.RP.2:** Understand the concept of a unit rate *a*/*b* associated with a ratio *a:b* with $b \neq 0$, and use rate language in the context of a ratio relationship. *For example, "This recipe has a ratio of 3 cups of flour to 4 cups of sugar, so there is 3/4 cup of flour for each cup of sugar." "We paid $75 for 15 hamburgers, which is a rate of $5 per hamburger."*	**6.1.2.4:** Use reasoning about multiplication and division to solve ratio and rate problems. *For example: If 5 items cost $3.75, and all items are the same price, then 1 item costs 75 cents, so 12 items cost $9.00.* **7.2.2.1:** Represent proportional relationships with tables,

concepts to solve proportion word problems. They represent information in the problems using a proportion schematic diagram and plan to solve the problem using various solution strategies (e.g., cross multiplication, equivalent fractions, unit rate). Common Core Standards: 6.RP.2, 6.RP.3.b, 7.RP.1, 7.RP.2.b MN State Standards: 6.1.2.4, 7.2.2.1, 7.2.2.2, 7.2.2.3, 7.2.4.2	**6.RP.3.b:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations.; Solve unit rate problems including those involving unit pricing and constant speed. *For example, if it took 7 hours to mow 4 lawns then at that rate, how many lawns could be mowed in 35 hours? At what rate were lawns being mowed?* **7.RP.1:** Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. *For example, if a person walks 1/2 mile in each 1/4 hour, compute the unit rate as the complex fraction 1/2/1/4 miles per hour, equivalently 2 miles per hour.* **7.RP.2.b:** Recognize and represent proportional relationships between quantities; Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.	verbal descriptions, symbols, equations and graphs; translate from one representation to another. Determine the unit rate (constant of proportionality or slope) given any of these representations. *For example: Larry drives 114 miles and uses 5 gallons of gasoline. Sue drives 300 miles and uses 11.5 gallons of gasoline. Use equations and graphs to compare fuel efficiency and to determine the costs of various trips.* **7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.2.3:** Use knowledge of proportions to assess the reasonableness of solutions. *For example: Recognize that it would be unreasonable for a cashier to request $200 if you purchase a $225 item at 25% off.* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*
Lesson 8 Scale Drawing Problems: Students will identify a proportional relationship in scale drawings and calculate dimensions in scale drawings using scale factors. They will represent	**6.RP.3:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. **7.RP.2.a:** Recognize and represent proportional relationships between quantities; Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.3.2.1:** Describe the properties of similarity, compare geometric figures for similarity, and determine scale

information in the scale drawing problems using a proportion diagram and plan to solve the problem. Common Core Standards: 6.RP.3, 7.RP.2.a, 7.G.1 MN State Standards: 7.2.2.2, 7.3.2.1, 7.3.2.2, 7.3.2.3	through the origin. **7.G.1:** Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	factors. **7.3.2.2:** Apply scale factors, length ratios and area ratios to determine side lengths and areas of similar geometric figures. **7.3.2.3:** Use proportions and ratios to solve problems involving scale drawings and conversions of measurement units.
Lesson 9 Scale Drawing Problems: Students will identify a proportional relationship in scale drawings and calculate dimensions in scale drawings/models using scale factors. They will represent information in the scale drawing/model problems using a proportion diagram and plan to solve the problem. Common Core Standards: 6.RP.3, 7.G.1 MN State Standards: 7.2.2.2, 7.2.2.3, 7.2.4.2, 7.3.2.3	**6.RP.3:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. **7.G.1:** Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.2.3:** Use knowledge of proportions to assess the reasonableness of solutions. *For example: Recognize that it would be unreasonable for a cashier to request $200 if you purchase a $225 item at 25% off.* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.* **7.3.2.3:** Use proportions and ratios to solve problems involving scale drawings and conversions of measurement units.
Lesson 10: Project: Recording Studio Students will demonstrate knowledge of material presented in lessons 1-9 in a real-world context.		

Standards included in lessons 1 – 9		
Lesson 11: Fractions, Percents and Decimals Students understand that percent is a special type of ratio and that fractions and percents are two ways to compare parts to a whole. They understand the relationship between fractions and percents and display the ability to convert between fractions, percents, and decimals. Common Core Standards: 6.RP.3.c, 7.EE.3 MN State Standards: 6.1.1.3, 6.1.1.4, 6.1.2.2, 7.4.2.1	**6.RP.3.c:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations; Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means 30/100 times the quantity); solve problems involving finding the whole, given a part and the percent. **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**6.1.1.3:** Understand that percent represents parts out of 100 and ratios to 100. **6.1.1.4:** Determine equivalences among fractions, decimals and percents; select among these representations to solve problems. **6.1.2.2** Apply the relationship between ratios, equivalent fractions and percents to solve problems in various contexts, including those involving mixtures and concentrations. *For example: If 5 cups of trail mix contains 2 cups of raisins, the ratio of raisins to trail mix is 2 to 5. This ratio corresponds to the fact that the raisins are 2/5of the total, or 40% of the total. And if one trail mix consists of 2 parts peanuts to 3 parts raisins, and another consists of 4 parts peanuts to 8 parts raisins, then the first mixture has a higher concentration of peanuts.* **7.4.2.1:** Use reasoning with proportions to display and interpret data in circle graphs (pie charts) and histograms. Choose the appropriate data display and know how to create the display using a spreadsheet or other graphing technology.
Lesson 12: Solving Percent Word Problems: Part-to-Whole Comparisons Students apply ratio concepts to solve percent problems. They represent information in the word problems using a ratio diagram and plan to solve the problem. Common Core Standards: 6.RP.3.c, 7.EE.3	**6.RP.3.c:** Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations; Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means 30/100 times the quantity); solve problems involving finding the whole, given a part and the percent. **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and	**6.1.3.3** Calculate the percent of a number and determine what percent one number is of another number to solve problems in various contexts. *For example: If John has $45 and spends $15, what percent of his money did he keep?* **7.1.2.5:** Use proportional reasoning to solve problems involving ratios in various contexts. *For example: A recipe calls for milk, flour and sugar in a ratio of 4:6:3 (this is how recipes are often given in large institutions, such as hospitals). How much flour and milk would be needed with 1 cup of sugar?* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the*

MN State Standards: 6.1.3.3, 7.1.2.5, 7.2.4.2	estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	*side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*
Lesson 13: Solving Percent Word Problems: Percent of Change Students apply ratio concepts to solve percent of change problems. They represent information in the word problems using a ratio diagram and change diagram and solve the problem. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.2.2.2, 7.2.4.2	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*

Lesson 14: Solving Percent Word Problems: Percent of Change Students apply ratio concepts to solve percent of change problems. They represent information in the word problems using a ratio diagram and change diagram and plan to solve the problem. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.2.2.2, 7.2.4.2	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*
Lesson 15: Percent of Change: Sales Tax and Tips Word Problems Students apply ratio concepts to solve sales tax and tips problems. They represent information in the word problems using ratio and change diagrams. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.2.2.2, 7.2.2.3	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems.* **7.2.2.3:** Use knowledge of proportions to assess the reasonableness of solutions. *For example: Recognize that it would be unreasonable for a cashier to request $200 if you purchase a $225 item at 25% off.*

Lesson 16: Solving Markup and Discount Word Problems: Percent of Change Students apply ratio concepts to solve markup and discount problems. They represent information in the word problems using the ratio and change diagrams and plan to solve the problem. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.2.2.2, 7.2.2.3, 7.2.4.2	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems* **7.2.2.3:** Use knowledge of proportions to assess the reasonableness of solutions. *For example: Recognize that it would be unreasonable for a cashier to request $200 if you purchase a $225 item at 25% off.* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*
Lesson 17: Solving Multistep Percentage Adjustment Problems Students solve problems involving multiple successive discounts and markups. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.2.2.2, 7.2.4.2	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	**7.2.2.2:** Solve multi-step problems involving proportional relationships in numerous contexts. *For example: Distance-time, percent increase or decrease, discounts, tips, unit pricing, lengths in similar geometric figures, and unit conversion when a conversion factor is given, including conversion between different measurement systems* **7.2.4.2:** Solve equations resulting from proportional relationships in various contexts. *For example: Given the side lengths of one triangle and one side length of a second triangle that is similar to the first, find the remaining side lengths of the second triangle. Another example: Determine the price of 12 yards of ribbon if 5 yards of ribbon cost $1.85.*

Lesson 18: Simple Interest Students calculate simple interest. Common Core Standards: 7.RP.3, 7.EE.3 MN State Standards: 7.1.2.4, 7.2.2.2, 7.2.4.2	**7.RP.3:** Use proportional relationships to solve multistep ratio and percent problems. *Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.* **7.EE.3:** Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. *For example: If a woman making $25 an hour gets a 10% raise, she will make an additional 1/10 of her salary an hour, or $2.50, for a new salary of $27.50. If you want to place a towel bar 9 3/4 inches long in the center of a door that is 27 1/2 inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.*	
Lesson 19: Review: Identifying and Representing Problem Types **Standards included in lessons 11 – 18** Students will review material from lessons 11-18 by identifying and categorizing word problems into appropriate problem types (e.g. ratio, proportion, percent and percent of change).		
Lesson 20: Review Project: IMAX Planetarium **Standards included in lessons 11-18** Students will demonstrate knowledge of material presented in Lessons 11-18 in a real-world context.		
Lesson 21: Review: Identifying and Representing Problem Types Standards included in lessons 1 – 18 Students will demonstrate understanding of unit content in a standardized test format.		